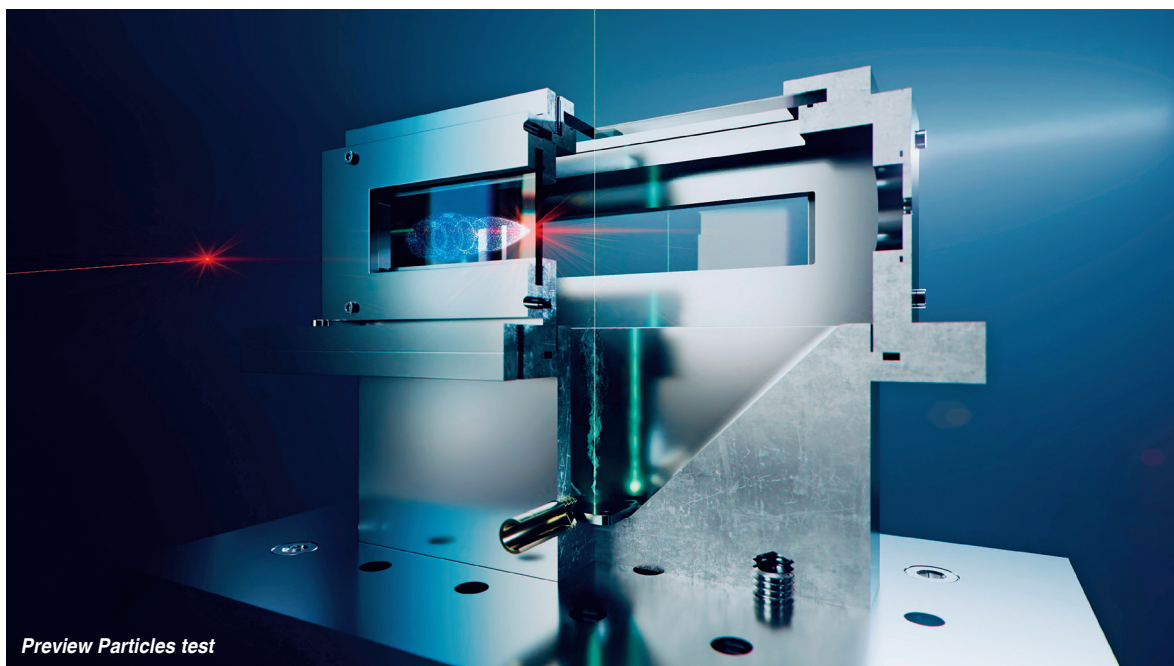


THE LITHOGRAPHY ARCHITECTURE Built For Multiple Future Generations

Making computer chips is becoming harder, slower, and more expensive. A new way of generating light could help the semiconductor industry move beyond some of today's most stubborn manufacturing limits.



Preview Particles test



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Modern chipmaking is increasingly an exercise in managing bottlenecks. Today's leading manufacturers rely on extreme ultra-violet (EUV) lithography, which uses 13.5nm light generated by firing powerful lasers at tiny droplets of tin. Once produced, this light must pass through a complex system of 11 mirrors before reaching the wafer. Each mirror reflects only about 70% of the light, meaning 30% is lost at every step. Because each reflection introduces loss, only a small fraction of the original light ultimately reaches the wafer surface. That reduced usable light directly affects wafer throughput and contributes to the high cost of advanced chip production. Think of it like trying to pass a bucket of water through 11 people in a line, where

some water spills at every handoff. By the time it reaches the end, only a small amount is left for the actual job.

Mirror loss, however, is only one of several constraints. EUV light generation itself remains inherently inefficient, while wafer materials require extremely precise exposure control: too little light prevents accurate pattern formation, too much can destroy the wafer. As feature sizes continue to shrink, optical systems also demand increasingly sophisticated mirror architectures and tighter tolerances, driving cost and engineering complexity even higher.

Because EUV operates at a fixed 13.5nm wavelength, further scaling cannot rely on changing the light itself. Instead, it requires increasing the system's numerical aperture (NA), which in turn